IMPLEMENTING SECURITY DEFENSES

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Security Policy

Definition :
A roadmap for securing systems and networks.

Key Points:

- Defines what needs to be secured (e.g., applications, passwords, connections).
- Provides rules for users and administrators (e.g., no password sharing, regular code reviews).

Example:
A policy might require port scans every six months on all external connections.

Importance:

- Without a policy, users and admins won’t know what’s allowed or required.
- Policies must be reviewed and updated regularly.

Vulnerability Assessment

Purpose:

Identify weaknesses in a system or network.

Methods:

- Penetration Testing: Simulating attacks to find vulnerabilities.
- Risk Assessment: Evaluating potential losses and odds of security incidents.

Tools:

Port scans, social engineering tests, and system checks.

Example:

Scanning for weak passwords or unauthorized programs in system directories.

Challenges:

- Vulnerability tools can also be used by attackers.
- Security through obscurity is not enough.

Intrusion Prevention

Definition:
Detecting and stopping unauthorized access attempts.

Techniques:

- Signature-Based Detection: Identifies known attack patterns (e.g., scanning for "/etc/passwd" in network packets).
- Anomaly Detection: Flags unusual behavior (e.g., abnormal login times or system calls).

Example:
A system might divert an attacker to a honeypot to monitor their activity.

Challenges:

- High false-alarm rates can reduce effectiveness.
- Balancing detection accuracy and usability.

Virus Protection

Threat:
Viruses can cause significant damage to systems.
Solutions:

- Antivirus Software: Scans for known virus patterns.
- Sandboxing: Runs suspicious code in a controlled environment.

Example:
A virus modifying itself to avoid detection can be caught by family-based pattern matching.
Best Practices:

- Avoid pirated software and unknown email attachments.
- Regularly update antivirus software.

Auditing, Accounting, and Logging

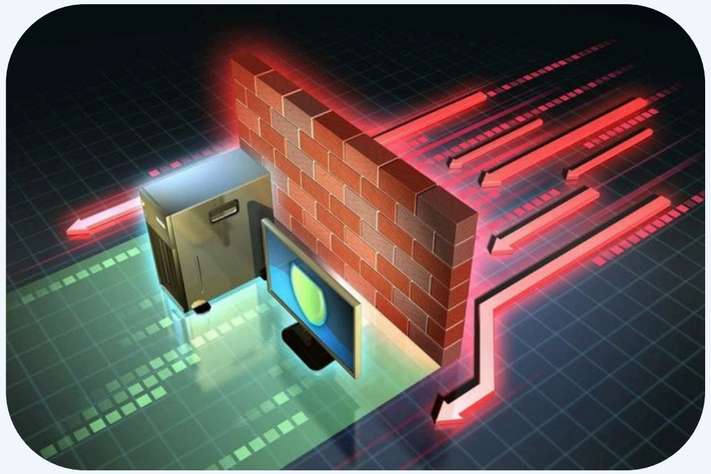
Purpose:
Monitor system activity for suspicious behavior.
Examples:

- Logging failed login attempts.
- Tracking system-call executions.

-**Example:**
Cliff Stoll detected an early UNIX break-in by analyzing accounting logs.
Benefits:

- Early detection of intrusions.
- Helps in forensic analysis after an attack.

Firewalls



Definition:
A barrier between trusted and untrusted networks.
Types:

- Network Firewalls: Filter traffic based on rules (e.g., allow HTTP, block FTP).
- Personal Firewalls: Protect individual devices.

Example:
A firewall might allow HTTP traffic to a web server but block FTP traffic.
Limitations:

- Cannot stop attacks within allowed protocols (e.g., HTTP buffer overflow).
- Vulnerable to spoofing and denial-of-service attacks.

Advanced Security Techniques

Address Space Layout Randomization (ASLR):

Randomizes memory addresses to make attacks harder.

Mobile OS Security:

- Separate read-only system partitions from user data.
- Use cryptographic hashing to detect tampering.

Example:

Android uses Linux’s dm-verity mechanism to hash system partitions.

Modern Hardware/Software:

Keep systems updated with the latest security patches.

THANK YOU !